

Lanthanides

This set is named after lanthanum, the first element in this series. The name “rare earth metals” is also given to these elements because they were discovered mixed together in complex minerals in Earth’s crust, and were thought to be uncommon. However, they are actually not rare but abundant. These metals – between barium (Ba) and hafnium (Hf) – should fit between the alkaline earth metals and the transition metals, but they are normally shown underneath the main table to save on space.



Atomic structure

Atoms of every element in this group have two outer electrons. The lanthanides have large atoms, all with six electron shells.



Physical properties

The lanthanides are dense, shiny metals, which tarnish easily when exposed to air. They do not conduct electricity very well.



Chemical properties

These elements react slowly with oxygen (O) at room temperature, but the reactions speed up when heated.



Compounds

Many lanthanides form compounds with oxygen called oxides. These are often used in lasers and magnets.